

Coursework Part [B]: [Soft Robotics]

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I. Q1 – CONTROLLING DIELECTRIC ELASTOMER ACTUATORS

(i) Derivation of the Thickness–Voltage Relation

At static equilibrium, the dielectric elastomer actuator (DEA) adopts the thickness t that minimises its total potential energy density Π . This gives the condition:

$$\frac{d\Pi}{dt} = 0. \quad (1)$$

The potential energy is composed of an elastic term from the incompressible neo-Hookean material model and an electrostatic term from the Maxwell stress, which promotes thickness reduction under voltage:

$$\Pi = \frac{\mu}{4} \left[\left(\frac{t}{t_0} \right)^4 + 2 \left(\frac{t_0}{t} \right)^2 - 3 \right] - \frac{\varepsilon_r \varepsilon_0}{2t^2} V^2. \quad (2)$$

The first bracketed term represents the stored strain energy associated with deviatoric deformation of an incompressible elastomer. The second term captures the electrical work performed by the applied field, which effectively compresses the film and competes with the mechanical stiffness.

Differentiating Π with respect to t yields

$$\frac{d\Pi}{dt} = \mu \left(\frac{t^3}{t_0^4} - \frac{t_0^2}{t^3} \right) + \frac{\varepsilon_r \varepsilon_0 V^2}{t^3}. \quad (3)$$

Setting this equal to zero gives the equilibrium condition. Multiplying by t^3 removes the negative powers and simplifies the expression:

$$\mu \left(\frac{t^6}{t_0^4} - t_0^2 \right) + \varepsilon_r \varepsilon_0 V^2 = 0. \quad (4)$$

Rearranging leads to

$$t^6 = t_0^6 \left(1 - \frac{\varepsilon_r \varepsilon_0}{\mu t_0^2} V^2 \right). \quad (5)$$

Taking the sixth root gives the closed-form thickness–voltage relationship:

$$t = t_0 (1 - KV^2)^{1/6}, \quad (6)$$

where

$$K = \frac{\varepsilon_r \varepsilon_0}{\mu t_0^2}. \quad (7)$$

The expression highlights two physically important effects. The thinning becomes highly nonlinear as V increases due to the $1/6$ exponent, and pull-in instability occurs as $(1 - KV^2) \rightarrow 0$, which corresponds to loss of mechanical equilibrium.

Since KV^2 must be dimensionless, K carries units of inverse voltage squared, ensuring consistency whether V is measured in volts or kilovolts.

(ii) Experimental Fit

The analytical model $t = t_0(1 - KV^2)^{1/6}$ was fitted to the data using nonlinear least squares. The initial thickness was set to $t_0 = 0.300$ mm (the measured value at $V = 0$). The optimisation yielded

$$K = 0.00131 \text{ kV}^{-2}, \quad R^2 = 0.999,$$

indicating an excellent match between the theoretical model and the measured behaviour.

The fitted value of K also determines the theoretical pull-in voltage:

$$V_{\text{lim}} = \frac{1}{\sqrt{K}} \approx 27.6 \text{ kV},$$

which is the point at which the denominator in the DEA thickness expression collapses and the film becomes unstable.

Figure 1 shows excellent agreement across the full voltage range, including the increasingly steep decline in t at high voltage, which arises naturally from the nonlinearity of the model.

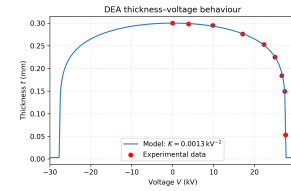


Fig. 1: DEA thickness–voltage behaviour and fitted model. The nonlinearity near pull-in is captured accurately.

(iii) Voltage Ranges for Precise and Rapid Actuation

A 0.05 mm stroke can be achieved in two distinct operating regimes, each corresponding to a different region of the $t(V)$ curve.

For precise regime (0–22.5 kV), the elastomer is relatively stiff and the thickness varies slowly with voltage. Using the derivative of the fitted model,

$$\left| \frac{dt}{dV} \right| \approx 1.5 \text{ } \mu\text{m/kV} \quad (V \approx 10 \text{ kV}),$$

indicating low sensitivity and therefore high robustness to voltage noise. This region is well suited for fine positioning where a smooth, predictable response is required.

For rapid regime (26.4–27.4 kV), near pull-in the electrostatic force dominates the mechanical restoring force. The

actuator becomes very compliant and the same stroke can be achieved with a ~ 1 kV change

$$\left| \frac{dt}{dV} \right| \approx 49 \text{ } \mu\text{m/kV} \quad (V \approx 27 \text{ kV}).$$

This region enables fast motion and high bandwidth but sacrifices stability and makes the system extremely sensitive to voltage disturbances.

(iv) Effect of Dielectric Breakdown Strength

The allowable electric field is restricted by the breakdown strength

$$E_{BD} = 124 \text{ kV/mm},$$

which must exceed the operating field $E = V/t$.

In the precise regime, even at maximum voltage:

$$E = \frac{22.5}{0.25} = 90 \text{ kV/mm} < E_{BD},$$

making this region safe.

In contrast, the rapid regime exceeds the breakdown limit immediately:

$$E = \frac{26.4}{0.20} = 132 \text{ kV/mm} > E_{BD}.$$

Thus the high-sensitivity regime is physically inaccessible for this material. While the theoretical model suggests the possibility of rapid actuation, dielectric breakdown prevents the system from entering this region. The DEA must therefore operate exclusively in the precise, low-strain regime to avoid electrical failure.

II. Q2 – ERROR SOURCES IN STRETCHABLE SENSORS

(i) Effect of Contact Resistance on Stretch Measurements

For a uniaxial resistive strain sensor, the resistance varies linearly with extension according to the Gauge Factor definition

$$R_{\text{ideal}}(\Delta L) = R_0 \left(1 + G \frac{\Delta L}{L_0} \right). \quad (8)$$

In practice, each terminal contributes a parasitic contact resistance R_{cp} , producing a measured resistance

$$R_m = R_{\text{ideal}} + 2R_{cp}. \quad (9)$$

Figure 2 shows the ideal and measured resistances for Sensors A, B, and C over $0 \leq \Delta L \leq 20$ mm. Because R_{cp} is constant, each measured curve is a vertical offset of its ideal counterpart; the slopes remain unchanged. Sensor B has the largest slope (dR/dL), hence the largest intrinsic sensitivity, while Sensor C shows the lowest due to its small gauge factor ($G = 0.6$).

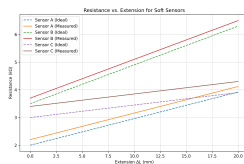


Fig. 2: Ideal and measured resistances for Sensors A–C. Contact resistance produces a constant vertical offset.

If the readout system incorrectly assumes $R_{cp} = 0$, it estimates extension by inverting the ideal model,

$$\Delta L_{\text{pred}} = \frac{(R_m - R_0)L_0}{R_0 G}. \quad (10)$$

Substituting $R_m = R_{\text{ideal}} + 2R_{cp}$ yields a constant positive bias independent of the true extension:

$$\Delta L_{\text{pred}} - \Delta L_{\text{true}} = \frac{2R_{cp}L_0}{R_0 G}, \quad (11)$$

Using the tabulated sensor parameters we can get the results as below:

Sensor A: Error ≈ 2.08 mm,

Sensor B: Error ≈ 1.43 mm,

Sensor C: Error ≈ 8.89 mm.

Sensor B is least affected by contact resistance, while Sensor C becomes effectively unusable without calibration due to its combination of low gauge factor and high contact resistance.

(ii) Quantisation Error and Sensor Selection

The measured resistance at $\Delta L = 10$ mm is

$$R_m = R_0 \left(1 + G \frac{10}{L_0} \right) + 2R_{cp}, \quad (12)$$

And the potential divider produces as follows, where $V_{\text{in}} = 5$ V and $R_k = 10$ k Ω :

$$V_{\text{out}} = \frac{R_k}{R_k + R_m} V_{\text{in}}, \quad M = \text{round} \left(31 \frac{V_{\text{out}}}{V_{\text{in}}} \right) \quad (13)$$

Using the provided parameters, we get the results:

Sensor	R_m (k Ω)	V_{out} (V)	M
A	3.16	3.80	24
B	5.10	3.31	21
C	3.85	3.61	22

To determine accuracy at $\Delta L = 10$ mm, we evaluate the total sensitivity of the measurement chain:

$$S = \left| \frac{dM}{d(\Delta L)} \right| = \underbrace{\frac{31}{V_{\text{in}}}}_{\text{ADC}} \cdot \underbrace{\left[\frac{-V_{\text{in}} R_k}{(R_k + R_m)^2} \right]}_{\text{Divider}} \cdot \underbrace{\left[\frac{R_0 G}{L_0} \right]}_{\text{Sensor}}. \quad (14)$$

The resulting sensitivities are:

$$S_A = 0.172 \text{ counts/mm},$$

$$S_B = \mathbf{0.190 \text{ counts/mm}},$$

$$S_C = 0.073 \text{ counts/mm}.$$

Although Sensor A has a higher gauge factor, Sensor B is the most accurate overall, because its operating resistance at 10 mm ($R_m \approx 5.1$ k Ω) lies closest to the fixed resistor $R_k = 10$ k Ω . A potential divider achieves maximal sensitivity when $R_m \approx R_k$. Sensor C is significantly limited by both low gauge factor and high contact resistance. **Therefore, Sensor B provides the highest measurement accuracy at $\Delta L = 10$ mm.**

III. Q3 – FIBRE OPTIC SENSORS

(i) Acceptance Angle vs. Surrounding Refractive Index

In a step-index PMMA optical fibre, light is guided by total internal reflection (TIR) at the core-medium boundary. The maximal incident angle α that can still undergo TIR is determined by the numerical aperture condition below, where $n_1 = 1.49$ is the fibre-core refractive index and n_2 is the refractive index of the surrounding medium.

$$\sin \alpha = \sqrt{n_1^2 - n_2^2}, \quad (15)$$

Equation (15) is only physically meaningful when the right-hand side does not exceed unity, since $\sin \alpha \leq 1$. For sufficiently low values of n_2 the quantity $\sqrt{n_1^2 - n_2^2}$ can become larger than 1; in practice the acceptance angle is then capped at $\alpha = 90^\circ$, corresponding to the limiting case where all incident rays are in principle accepted by the fibre. Conversely, as n_2 increases towards n_1 , the refractive-index contrast decreases, the right-hand side tends towards zero, and the acceptance angle shrinks rapidly.

Figure 3 plots α for $1.0 \leq n_2 \leq 1.5$. The three required reference materials—air, silicone rubber, and fused silica—are marked explicitly on the curve. A horizontal reference line at $\alpha = 30^\circ$ indicates the illumination cone of the light source considered in part (ii).

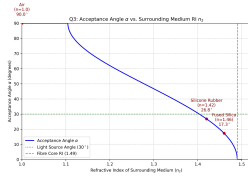


Fig. 3: Acceptance angle α versus surrounding medium refractive index n_2 , computed using Eq. (15). The dashed horizontal line marks the 30° illumination limit. Air achieves the maximal angle (90°), while silicone rubber ($n_2 = 1.42$) and fused silica ($n_2 = 1.46$) yield much smaller acceptance angles.

As seen in the figure, the acceptance angle decreases monotonically with increasing n_2 . For silicone rubber ($n_2 = 1.42$), $\alpha \approx 26.8^\circ$, already below the 30° threshold; fused silica ($n_2 = 1.46$) falls even further to $\alpha \approx 17.3^\circ$. The only material among the highlighted examples that meets the $\alpha \geq 30^\circ$ requirement is air.

(ii) Material Exclusion Based on 30° Illumination Cone

To fully couple a light source whose rays extend up to a maximal incident angle $\theta_{\max} = 30^\circ$, the fibre must have an acceptance angle satisfying:

$$\alpha \geq 30^\circ \iff \sqrt{n_1^2 - n_2^2} \geq \sin 30^\circ = 0.5. \quad (16)$$

Solving for n_2 gives the critical refractive index:

$$n_2 \leq \sqrt{n_1^2 - 0.5^2} = \sqrt{1.49^2 - 0.25} \approx 1.4036. \quad (17)$$

Any medium with $n_2 > 1.404$ produces $\alpha < 30^\circ$ and therefore cannot support complete coupling of the illumination cone.

Using refractive-index measurements for soft contact lens materials reported in Kim *et al.* (2020), each material was classified according to this threshold. Table I summarises the compatibility.

TABLE I: Compatibility of soft materials (Kim *et al.*, 2020) with the 30° acceptance-angle requirement. Materials with refractive index $n_2 > 1.404$ are excluded because they yield $\alpha < 30^\circ$.

Material	Refractive Index	Status
Biotrue ONEday	1.374	OK
Dailies Aqua Comfort Plus	1.380	OK
Proclear	1.390	OK
Acuvue 2	1.400	OK
Clariti 1-Day	1.4003	OK
Biofinity	1.406	EXCLUDED
Acuvue TrueEye 1-Day	1.410	EXCLUDED
PureVision 2	1.426	EXCLUDED
Air Optix Aqua	1.420	EXCLUDED
Air Optix Night & Day	1.430	EXCLUDED
Premio	1.423	EXCLUDED
Acuvue Oasys	1.420	EXCLUDED
Dailies Total 1	1.420	EXCLUDED
Total Excluded	—	8

The result is clear that only materials with refractive index below approximately 1.40 maintain sufficient numerical aperture to capture a 30° illumination cone. Most silicone-hydrogel and silicone-rubber materials fall above this threshold and therefore cannot be used in this fibre-optic sensing configuration. Low-RI hydrogels, however, remain compatible and provide adequate optical contrast for robust coupling.

IV. Q4 – FINITE-ELEMENT APPROXIMATION OF SOFT MATERIALS

(i) Model Construction

To approximate the dynamics of a soft rod, a lumped mass-spring chain with four rigid cubes and three linear springs was used (Fig. 4). Each cube has mass $m = 1$ kg; the springs have stiffness $k = 10$ N/m and natural length 0.4 m, while the initial distance between cube centres is 0.5 m, so each spring is slightly pre-stretched at $t = 0$.

A 10 N rectangular force pulse of duration 0.1 s was applied to Cube 4 to emulate an impulsive actuation at the distal end. The motion was simulated for 10 s to observe wave propagation, deformation and rebound along the chain.

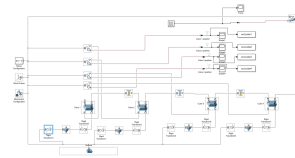


Fig. 4: System Structure

(ii) Dynamic Response and Stiffness Effects

Wave propagation through the chain

Figure 5 illustrates the temporal response of all four cubes for three stiffness values: $k = 1, 10$, and 100 N/m . Immediately after the force pulse is applied, Cube 4 responds first, followed by Cube 3, Cube 2, and finally Cube 1. This ordered activation clearly demonstrates finite-speed wave propagation, typical of compliant soft materials.

The softest chain ($k = 1 \text{ N/m}$) exhibits a significant phase delay: Cube 1 remains almost stationary for the first two seconds, indicating that the mechanical wave requires substantial time to traverse the compliant structure. The high-compliance case also shows the largest relative deflections between cubes, representing substantial internal strain.

In contrast, the stiff system ($k = 100 \text{ N/m}$) produces tightly grouped trajectories. All cubes respond within a short time interval, consistent with a high wave speed and behaviour approaching that of a rigid body.

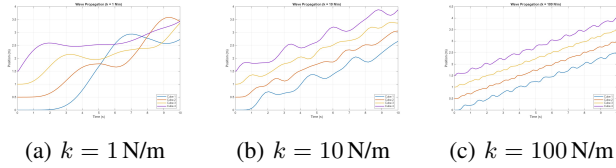


Fig. 5: Wave propagation along the cube chain for different stiffness values

Stiffness-dependent oscillatory behaviour

A more detailed comparison is provided in Fig. 6, showing the individual trajectories of each cube for the three stiffness values.

Because the natural frequency is proportional to $\omega_n = \sqrt{k/m}$, higher stiffness leads to faster oscillations. This is clearly visible in the $k = 100 \text{ N/m}$ case, where high-frequency ripples appear on top of the overall forward motion. The softest system ($k = 1 \text{ N/m}$), in contrast, demonstrates slow, large-amplitude motion resembling sloshing, with pronounced relative displacement between cubes.

Since no damping elements were present in this part of the model, none of the oscillations decay; instead, energy is continually exchanged among the masses and springs, consistent with a conservative elastic system.

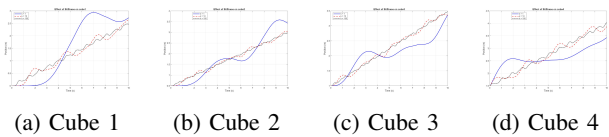


Fig. 6: Dynamic trajectories of individual cubes for varying stiffness k

(iii) Energy Dissipation and Damping Behaviour

To obtain realistic soft-material dynamics, viscous damping elements were added in parallel with each spring ($F_c = -c\dot{x}$). The stiffness was fixed at $k = 10 \text{ N/m}$, while the damping coefficient was varied as $c = 0, 1$, and 10 Ns/m to

study the effects of underdamping, moderate damping, and overdamping.

Global system behaviour

The system-wide responses are shown in Fig. 7. With no damping ($c = 0$), oscillations propagate indefinitely due to continuous energy reflection. For moderate damping ($c = 1$), oscillations still appear but decay over time, revealing a clean travelling wave from Cube 4 to Cube 1 that eventually settles. This behaviour is representative of soft robotic actuators in practice, where motion must be transmitted while avoiding excessive vibration.

For strong damping ($c = 10$), the disturbance becomes severely attenuated. Cube 1 and Cube 2 exhibit substantial phase lag and reduced amplitude, reflecting the low-pass filtering nature of overdamped viscoelastic materials.

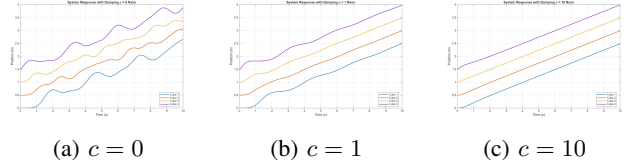


Fig. 7: Effect of damping on global wave propagation.

Local element responses

Figure 8 shows how vibration amplitude is reduced as the wave travels through the chain. Cube 4 (excitation point) shows a large overshoot for $c = 1$, while strong damping removes overshoot but slows the response. Cubes 3 and 2 exhibit progressive attenuation, demonstrating the distributed nature of viscoelastic damping. At Cube 1, the overdamped system moves only minimally, indicating that excessive damping isolates the distal end-effector from the input actuation.

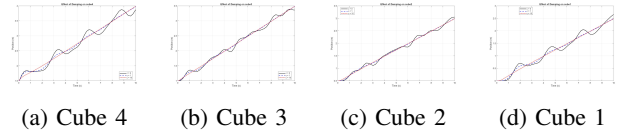


Fig. 8: Local damping response for each cube

In conclusion, the finite-element chain captures key behaviours of soft materials, including finite-speed wave propagation, stiffness-dependent deformation, and damping-induced attenuation. Among the tested parameters, moderate damping ($c = 1 \text{ Ns/m}$) provides the most useful compromise between stability and transmissibility for soft robotic applications.